

RESUME

PERSONAL PROFILE

Name: Muhammad Iqbal, Shamreed

Address: Mulki, Mangalore, PIN: 574154, Karnataka, INDIA

Contact Number: +91 9686581162

Date of Birth: 22/08/2004

Place of Birth: Surathkal, Karnataka

Email: shamreed02@gmail.com

LinkedIn: <https://www.linkedin.com/in/shamreedali/>

GitHub: <https://github.com/Shamreed>

Portfolio: <https://shamreed.github.io/Shamreed-portfolio/>



EDUCATION:

2022- 2026 Bachelor of Technology in Artificial Intelligence and Machine Learning
NMAM Institute of Technology, Nitte, Karkala, India
Affiliated to Nitte University.

CGPA: 8.66

PROJECTS:

Pothole Detection and Severity Analysis | Python, YOLOv8, PyTorch, Django, React

- Developed a deep learning - based pothole detection and severity analysis system using YOLOv8 on dashcam video data.
- Built an end-to-end processing pipeline integrating OCR, GPS data extraction, and database storage for geospatial pothole mapping.
- Developed a Django-React web application to visualize detected potholes on an interactive map, enabling efficient monitoring and analysis.

Gear-Driven Walking Robot | Mechanical Design, Gears, Linkages, Motors, Robotics, Hardware Integration

- Designed and built a compact walking robot using interconnected gears, mechanical linkages, and a motor-driven locomotion mechanism.
- Developed and assembled a gear-based actuation system that converts motor rotation into coordinated leg movement for stable walking motion.
- Integrated a rechargeable battery and solar panel for dual-power operation, and tested the mechanical system for reliable locomotion and overall performance.

Alzheimer's Disease Classification Using Deep Learning | Python, TensorFlow, Keras, Deep Learning, Transfer Learning, CNNs

- Developed a deep learning-based image classification system for Alzheimer's disease classification.
- Trained and evaluated multiple CNN architectures, including DenseNet169, VGG16, ResNet50, and EfficientNetB0, using transfer learning.
- Conducted model comparison and performance evaluation using accuracy, precision, recall, F1-score, and ROC-AUC to identify the most effective architecture.

TECHNICAL SKILLS:

- **Programming Languages:** Python, C++, Java, JavaScript, SQL
- **Machine Learning & Deep Learning:** Machine Learning, Deep Learning, Neural Networks, Transfer Learning, Model Training & Fine-Tuning, Model Evaluation, Reinforcement Learning, PyTorch, TensorFlow, Keras, scikit-learn
- **Multimodal AI & Robotics:** Multimodal AI, ROS 2 (Basic), Arduino, Hardware Integration
- **Software Engineering:** Git, GitHub, Docker, REST APIs, Spring Boot
- **Databases & Web:** MySQL, React.js, HTML, CSS
- **Hardware & Manufacturing:** Soldering, Basic Electrical Wiring, Quality Inspection

INTERNSHIPS:

Software Development Intern - Jun 2025 - Aug 2025

IHUB IT Solutions Mangalore, Karnataka

- Developed backend services for a full-stack Blog platform using Java Spring Boot, implementing secure REST APIs and role-based authentication.
- Integrated MySQL databases and implemented role-based authentication to support secure and scalable application workflows.
- Followed Git-based software engineering practices, contributing to debugging, testing, code maintainability, and reliable system development.

LANGUAGE PROFICIENCY:

- Japanese Language (N5): (Completed NPTEL/SWAYAM course)
- English: Full professional proficiency
- Hindi: Full professional proficiency
- Kannada: Full professional proficiency
- Tulu: Full professional proficiency
- Byari: Full professional proficiency

ACHIEVEMENTS:

- **2nd Place – Final Year Major Project Expo:** Recognized for the development of the *Pothole Detection and Severity Analysis* system, a multimodal AI solution integrating computer vision, depth estimation, and sensor data.
- **Certificate of Presentation – 7th International Conference on Computational Vision and Bio-Inspired Computing (ICCVBIC 2026):** Presented the *Pothole Detection and Severity Analysis* project at an international technical conference.

CERTIFICATIONS:

- Introduction to Japanese Language and Culture – NPTEL (2025)
- Agile Scrum in Practice – Infosys (2024)
- Programming in Python – NMAMIT (2024)
- Time Management – Infosys (2024)